

Stony Brook University
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ASSIGNMENT 3: Utility Maximization and Choice

Due: March 4, in class.

Instructions: Show all your work or points will be deducted.

1. In this problem, we will explore an application of the consumer choice model to show how flexible it is.

Consider the following scenario. You have two midterm exams upcoming, and you have to decide how to allocate your time. After eating, sleeping, exercising and maintaining some level of social life, you have 10 hours each day in which to study for your exams, i.e. 10 hours is your *total amount of study time*. You have figured out that your GPA from your two courses, Intermediate Microeconomic Theory and Biology takes the following form

$$GPA = \frac{4}{7} (2\sqrt{E} + \sqrt{B}),$$

where E is the number of hours you devote to Economics and B are the hours you devote to Biology, which should be regarded as continuous variables. Suppose further that you only care about your GPA.

- (a) Setup the constrained maximization problem and solve for the optimal allocation of study time using the Lagrangian method.
 - (b) If you follow the optimal strategy you found in (a), what will be your GPA?
 - (c) The Lagrange multiplier λ has a special meaning in this problem: it is the value, measured in GPA units of increasing your total amount of study time by one more hour. Compute this value by calculating the optimal λ , i.e. solving for it using the first order conditions.
 - (d) Now suppose that Intermediate Microeconomic Theory is a very hard class, so for every hour that you study of it, you need to take a one hour break, i.e. if you study for E hours Economics, the actual time that you need to spend is $2E$. What is your optimal allocation of study time in this case? What is the value of increasing your total amount of study time by one hour?
2. Consider the following utility function (referred as a quasi-linear utility function as it is linear in the second element):

$$u(x, y) = \sqrt{x} + y,$$

with prices and income given by: $p_y = 1$, $p_x \in \mathbb{R}_+$ and $m \in \mathbb{R}_+$.

- (a) For the special case of $p_x = \frac{1}{8}$ and $m = 1$, show analytically and graphically that there is no positive consumption choice $(x^*, y^*) \geq 0$ which satisfies the tangency condition:

$$\frac{u_x(x, y)}{u_y(x, y)} = \frac{p_x}{p_y}.$$

Briefly describe where the failure of the tangency condition stems from and what does it mean. *Hint: Setup the tangency condition and solve it. What do you get? Is it feasible?*

- (b) If we insist that $(x^*, y^*) \geq 0$, then describe, analytically and graphically, the optimal consumption choice for $p_x = \frac{1}{8}$, $m = 1$. *Hint: Think about corner solutions.*
- (c) Now suppose that $p_x \in \mathbb{R}_+$ and $m \in \mathbb{R}_+$ are unspecified. Suppose that we *don't insist on positive values* of either x or y . For which values of p_x and m do you get a nonsensical solution in the sense that $y^* < 0$?

3. Mr. Bond derives utility from martinis (m) in proportion to the number he drinks:

$$U(m) = m.$$

However, Mr. Bond is very particular about his martinis: He only enjoys them made in the exact proportion of two parts gin (g) to one part vermouth (v). Hence, we can rewrite Mr. Bond's utility function as

$$U(m) = U(g, v) = \min\left(\frac{g}{2}, v\right).$$

- (a) Graph Mr. Bond's indifference curve in terms of g and v for various levels of utility. Show graphically that regardless of the prices of the two ingredients, Mr. Bond will never alter the way he mixes martinis.
- (b) Calculate the demand functions for g and v . Notice that because the problem involves a perfect complements utility function, you cannot solve for utility-maximizing decisions by using calculus.
4. Mr. Bear only consumes corn chips and cookies. Mr. Bear's utility function is $u(x_1, x_2) = \min\{x_1 + 2x_2, 2x_1 + x_2\}$ where x_1 is his consumption of corn chips and x_2 is his consumption of cookies.
- (a) Graph Mr. Bear's indifference curves for different utility levels.
- (b) Suppose that the price of corn chips is $p_1 = 3$ and the price of cookies is $p_2 = 2$, and income is $m = 10$. How many units of corn chips and how many units of cookies should he consume to maximize his utility subject to his budget?
- (c) Now suppose that the price of corn chips is $p_1 = 1$ and the price of cookies is $p_2 = 3$, and Mr. Bear's income is 6. How many units of corn chips and how many units of cookies will Harry consume?
- (d) At what prices will Mr. Bear consume only corn chips and no cookies?
- (e) At what prices will he consume only cookies and no corn chips?
- (f) At what price-income combinations does Mr. Bear choose to consume equal quantities of corn chips and cookies?
5. Camilo gets m dollars per week for lunch money. He likes to consume for lunch sandwiches (s) and coke (c). His preferences for both goods can be described by the utility function $u(s, c) = s^{0.7}c^{0.3}$. The price of coke is p_c . However, Camilo likes to prepare his own sandwiches. He likes *triple-decker sandwiches*: he combines four slices of bread (b) with three slices of ham (h), so that each slice forms a "deck". Thus, s sandwiches can be created according to the relation $s = \min\{3b, 4h\}$. The price of bread is denoted as p_b and the price of a slice of ham is denoted as p_h . Thus Camilo's preferences in terms of the three goods (b, h, c) can be described as

$$u(b, h, c) = (\min\{3b, 4h\})^{0.7} c^{0.3}.$$

Although Camilo's problem involves three goods, in this exercise you will see that you can use the examples and techniques we discussed in class for two-goods utility functions to solve this problem.

- (a) State (but do not solve) Camilo's utility maximization problem with respect to the three goods subject to his budget constraint.
- (b) Define p_s as the *unit price of a sandwich*, i.e. the cost of making one sandwich taking into consideration the cost of all its components. What is p_s equal to?
- (c) We can simplify Camilo's problem considerably by considering sandwiches as a *composite good*, i.e. a good that is composed of several others. Compute the optimal choices of sandwiches and coke using Camilo's *reduced problem*: $\max u(s, c)$ subject to $p_s s + p_c c = m$.
- (d) What proportion of Camilo's income is used for sandwiches? Call this proportion y .
- (e) Given y , solve for Camilo's optimal choices of slices of bread and ham by solving the problem $\max s = \min\{3b, 4h\}$ subject to $p_b b + p_h h = y$.
- (f) Argue that the procedure in items (b)–(e) is equivalent to solving the problem you defined in (a).